

Multi-Layer Perceptron Based Health Risk Prediction for Lung and Other Diseases

Project submitted in partial fulfilment for the requirement of the degree of
Bachelor of Technology in Computer Science and Engineering

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Declaration

We hereby declare that the project titled “**Multi-Layer Perceptron Based Health Risk Prediction for Lung and Other Diseases**” submitted to Veer Surendra Sai University of Technology for the award of the degree of Bachelor of Technology in Computer Science and Engineering & Information Technology is a result of original work carried out in this report. We also declare that the work has not been submitted, in whole or in part, to any other university as an exercise for a degree or any other qualification.

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Abstract

Accurate prediction of critical health conditions like lung cancer, jaundice, and heart problems is essential for timely diagnosis and effective intervention. In this study, we propose a machine learning-based approach for health risk prediction using a survey dataset of 500 samples. We developed and evaluated multiple models, including Multi-Layer Perceptron (MLP), Random Forest, and Long Short-Term Memory (LSTM), for lung cancer prediction. The MLP model achieved the highest accuracy of 0.83, improving to 0.84 with engineered features such as risk score, smoke-alcohol interaction, age group, physical symptom count, and mental health score, and further to 0.89 with reduced feature selection. The relationship between lung cancer and mental health score was also examined, with Random Forest achieving 78% and LSTM 64% accuracy. A new feature “jaundice” was derived from alcohol consumption, chronic disease, and yellow fingers, revealing its association with lung cancer (Random Forest: 82%, MLP: 69%). Additionally, a heart problem indicator was constructed from existing symptoms, showing a strong relation with lung cancer (Random Forest and MLP: 93%). While the model performance is promising, future work will focus on addressing hereditary dependencies in heart conditions and identifying specific organ-level impacts. This study demonstrates the potential of MLP-based approaches for efficient, multi-disease health risk prediction using survey data.

Keywords: Health risk prediction, Lung cancer, Jaundice, Heart problem, Multi-Layer Perceptron, Random Forest, LSTM, Feature engineering, Survey data, Mental health.

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1. Introduction

The global burden of respiratory and lifestyle-related diseases such as lung cancer, heart disease, and mental health disorders continues to rise due to factors including pollution, smoking, stress, and sedentary lifestyles. Lung cancer is still a top cause of global cancer deaths, and early diagnosis is key to boosting survival chances. Traditional diagnostic methods often involve clinical assessments, imaging, and biopsy, which can be time-consuming, costly, and invasive. In response, machine and deep learning are increasingly used in medical diagnostics for fast, non-invasive predictions.

This study explores the development of predictive models using structured health data encompassing features such as smoking habits, anxiety, peer pressure, fatigue, and other physiological symptoms. Machine learning models including Random Forest and Multi-Layer Perceptron (MLP), along with sequence-based models like Long Short-Term Memory (LSTM), are applied to predict health outcomes such as lung cancer, mental health score, jaundice, and potential heart problems. Feature engineering techniques are also used to derive new predictors, such as composite risk scores and health-related indices, enhancing model performance and interpretability.

1.1. Motivation

The motivation behind this project stems from the need for accurate, early, and non-invasive detection of chronic health risks through computational models. Early-stage symptoms of diseases like lung cancer or jaundice are often subtle and underreported. Moreover, mental health conditions frequently go undetected due to stigma or lack of clinical access. Traditional diagnostic systems are limited by the availability of healthcare infrastructure and medical professionals, especially in resource-constrained regions.

Given this context, integrating machine learning into healthcare offers a scalable and efficient solution for health risk prediction. With accessible datasets comprising personal, behavioral, and physiological features, predictive models can help identify high-risk individuals for timely intervention. By examining the interrelation between physical symptoms, lifestyle behaviors, and health outcomes, this project aims to support the development of digital tools that augment clinical decision-making.

Additionally, introducing new features such as mental health scores and composite risk indicators allows for a more holistic understanding of patient health. The project's findings indicate that optimized models such as MLP, when used with engineered features, can significantly outperform traditional classifiers, emphasizing the value of data preprocessing and feature enhancement in medical AI applications.

1.2. Objective

The primary objective of this project is to develop a comprehensive machine learning-based system for early prediction and assessment of lung cancer and related health risks using structured patient data. This study leverages traditional and advanced machine learning techniques, including Multi-Layer Perceptron (MLP), Random Forest, and Long Short-Term Memory (LSTM) networks, to classify individuals based on health conditions such as lung cancer, jaundice, and potential heart problems.

The research incorporates novel feature engineering methods to derive new indicators like risk score, smoke-alcohol interaction, physical symptom count, and mental health score, thereby improving predictive accuracy. Among the models tested, MLP achieved the highest accuracy (up to 0.89), demonstrating the effectiveness of combining domain-driven features with neural network-based classifiers.

Furthermore, the project explores interconnections between lung cancer and secondary conditions like mental health deterioration and jaundice, uncovering clinically significant patterns. The aim is to present an interpretable, efficient, and scalable health prediction framework suitable for aiding early diagnosis and preventive healthcare strategies.

2. Literature Review

In recent years, machine learning (ML) and deep learning (DL) techniques integration has transformed the landscape of medical diagnostics, offering data-driven solutions for early disease detection. A broad range of studies have demonstrated how ML models can effectively predict critical health conditions, including lung cancer, heart disease, jaundice, and mental health disorders, using clinical and behavioral data.

2.1. Lung Cancer Prediction

Lung cancer is a major global killer, and catching it early is vital for better survival. Sharma and Goyal [1] demonstrated the effectiveness of ML algorithms such as SVM and Random Forest in early-stage lung cancer prediction. Similarly, Rahman and Muniyandi [6] compared data mining techniques, identifying their utility in detecting lung cancer from structured health records. Choudhury and Gupta [12] offered a broader review of ML-based lung cancer detection methods, highlighting improvements in accuracy through feature engineering. Feature selection is also emphasized in the work by Hosseini et al. [17], where selection methods were applied to enhance model performance.

LSTM models, although traditionally used for sequential data, have been applied to cancer prediction with moderate success [8][9]. They are often combined with neural network enhancements such as Dropout [10] to prevent overfitting and improve generalizability in small datasets.

2.2. Heart Disease Detection

The prediction of heart-related conditions through ML has also gained traction. Mohan et al. [11] utilized hybrid ML techniques to achieve high accuracy in heart disease detection. Dwivedi [13] conducted a comparative evaluation of algorithms, concluding that ensemble-based methods like Random Forest and gradient boosting perform effectively in complex medical datasets. The success of XGBoost in predictive modeling has been affirmed by Chen and Guestrin [14], who developed it as a scalable and high-performance algorithm for classification tasks.

The present study also investigates the correlation between lung cancer and heart disease. Such comorbid relationships are vital for understanding overlapping risk profiles, and integrating them into ML models helps generate more comprehensive predictions.

2.3. Mental Health and Comorbidity Analysis

Mental health is increasingly recognized as a factor influencing physical diseases. Haraty and Zantout [15] applied ML algorithms to predict mental health issues, while Zhang et al. [3] emphasized the psychological impact on chronic illnesses, establishing a data-driven link between mental and physical health. Including mental health scores in lung cancer models provides a more holistic view, improving predictive accuracy and real-world applicability.

The integration of behavioral and environmental features, such as smoking, alcohol use, and chronic illness, can uncover hidden correlations with secondary conditions like jaundice. Rajalakshmi and Ganesan [18] explored jaundice prediction using supervised learning algorithms, and Jain et al. [19] discussed chronic disease prediction using multiple ML models.

2.4. Deep Learning in Medical Imaging and Classification

Deep neural networks have been extensively used in medical image classification. Esteva et al. [3] achieved dermatologist-level performance in skin cancer detection using CNNs, while Minaee et al. [2] reviewed COVID-19 detection through image-based DL techniques. These studies confirm the robustness of deep architectures in feature learning from complex data representations. The foundational work by Goodfellow et al. [5] underpins much of today's DL approaches in medical applications. Although image data was not the focus of this study, the architecture, and principles of deep learning, particularly Multi-Layer Perceptron (MLP), remain highly relevant. Dropout regularization [10] and backpropagation techniques further enhance performance in noisy or reduced data scenarios.

2.5. Optimization and Feature Engineering

Performance improvements through optimization and engineered features have been widely studied. Saqlain et al. introduced domain-driven features like risk scores to improve classification tasks, while King and Zeng [16] addressed rare event prediction using logistic regression. These works align with the present study's creation of new composite features such as "mental health score" and "physical symptoms count".

Abdar et al. [20] developed a hybrid logistic regression and Naive Bayes model to detect liver disease, showing how combining classifiers can handle imbalanced and noisy health data. Feature selection and integration remain essential in boosting the predictive power of models across different medical conditions.

2.6. Summary

The literature strongly supports the use of machine learning (ML) and deep learning (DL) models such as Multi-Layer Perceptron (MLP), Random Forest, and Long Short-Term Memory (LSTM) in the early prediction of complex diseases like lung cancer, heart conditions, and comorbidities. These models have proven effective in processing health data and identifying patterns that aid in timely diagnosis. Recent studies emphasize the importance of feature engineering—such as incorporating risk scores, age groups, and symptom counts—as well as integrating mental health metrics to improve model accuracy and reflect real-world health complexities.

This project builds upon these findings by developing a comprehensive, multi-disease prediction model using an enhanced MLP architecture. Through optimized feature selection and reduced datasets, the model achieves high prediction accuracy across multiple conditions, including lung cancer, jaundice, and heart problems. By aligning with current trends in model refinement and mental-physical health integration, this work contributes to the growing field of data-driven public health solutions.

3. Proposed Model

This study utilizes a health-related dataset comprising 516 instances with multiple binary and categorical features related to respiratory and lifestyle conditions. The primary goal was to prepare this dataset for training predictive models by performing targeted preprocessing and feature engineering to enhance model accuracy and interpretability.

Table 1: Original Dataset

GEN DER	AG E	SMOK ING	YELLO W_FIN GERS	AN XIE TY	PEER_ PRESS URE	CHRONI C DISEASE	FAT IG UE	ALL ER GY	WHE EZIN G	ALCOHOL CONSUMI NG	COUG HING	SHORTNE SS OF BREATH	SWALLOWI NG DIFFICULTY	CHE ST PAIN	LUNG _CAN CER
M	69	1	2	2	1	1	2	1	2	2	2	2	2	2	YES
M	74	2	1	1	1	2	2	2	1	1	1	2	2	2	YES
F	59	1	1	1	2	1	2	1	2	1	2	2	1	2	NO
M	63	2	2	2	1	1	1	1	1	2	1	1	2	2	NO
F	63	1	2	1	1	1	1	1	2	1	2	2	1	1	NO

3.1. Data Engineering and Feature Engineering

Initial preprocessing included label encoding of categorical attributes such as gender and disease status. All binary features encoded with 1 (No) and 2 (Yes) were standardized to 0 and 1 respectively for consistency. New synthetic features were generated to better capture interrelated health risk factors. A Risk Score was calculated by summing critical risk indicators like allergy, wheezing, alcohol consumption, coughing, and swallowing difficulty. Additionally, a Smoke-Alcohol interaction feature was introduced to capture the combined effect of smoking and alcohol consumption.

Age was transformed into a categorical Age Group variable using bins to represent youth, middle-aged, and elderly. A Physical Symptoms Count was derived from symptoms such as coughing, chest pain, shortness of breath, and swallowing difficulty, while a Mental Health Score combined anxiety and peer pressure indicators.

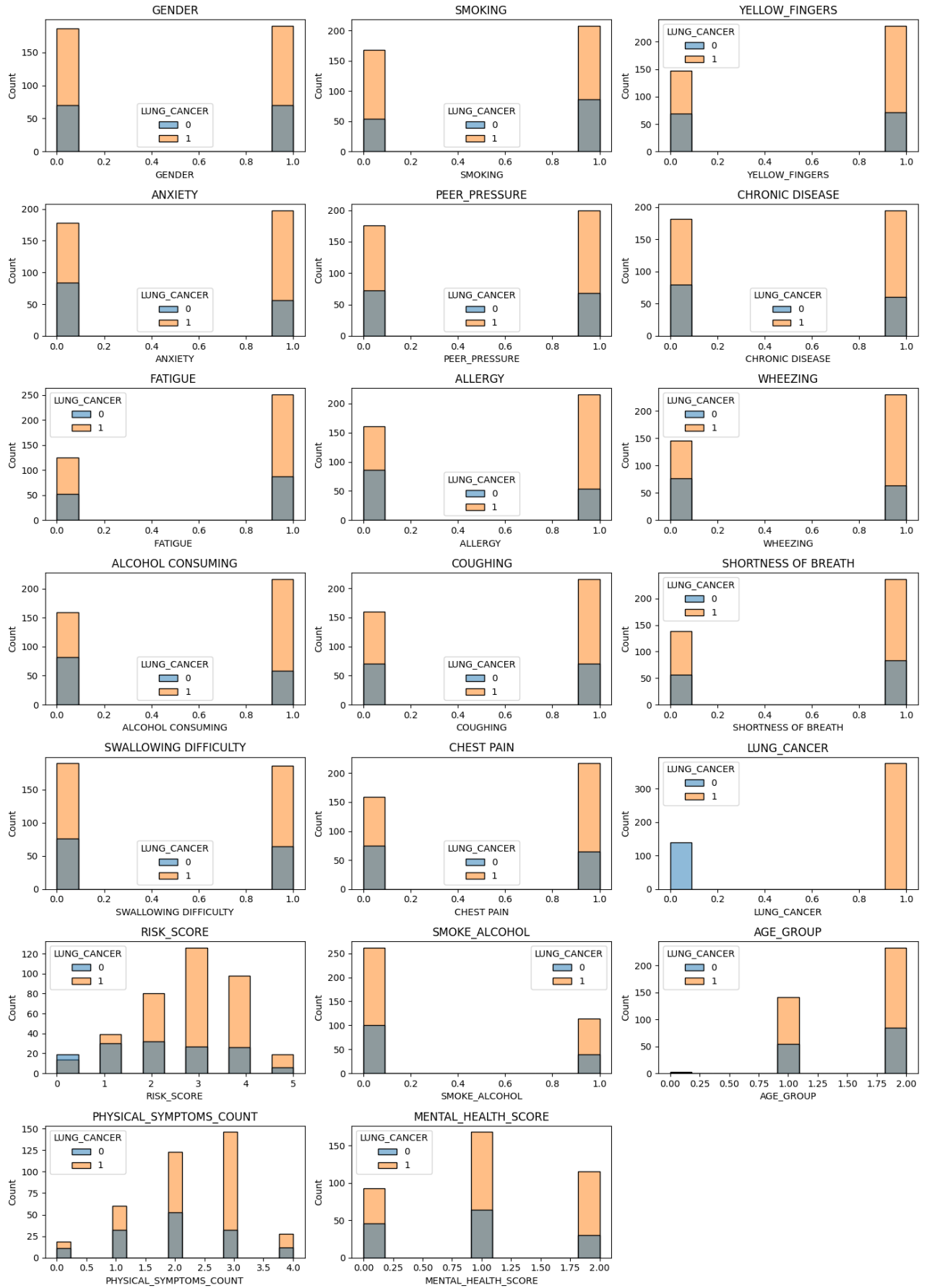


Fig. 1: Distribution of Original and Synthetic Features Stratified by Lung Cancer Status

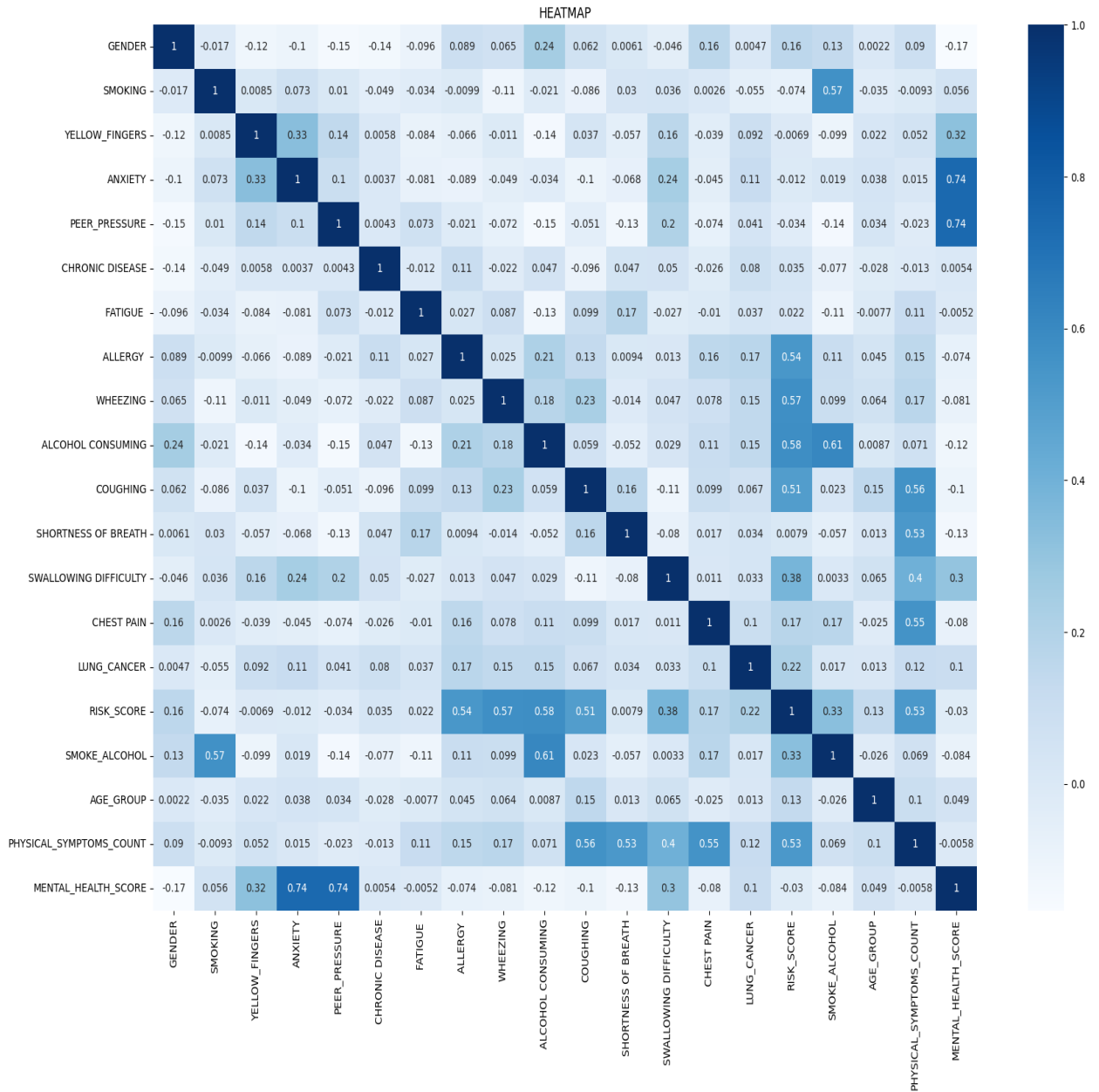


Fig. 2: Heatmap for Lung Cancer Feature Correlation Analysis

To focus on mental health associations, a reduced dataset excluding the original mental health columns was created. Furthermore, a Jaundice feature was synthesized based on alcohol consumption, chronic disease, and yellow fingers, marking presence when at least two conditions co-occurred. Lastly, a Heart Problem label was derived from a cumulative score of multiple risk indicators, with a threshold used to determine likely cardiac involvement. These engineered datasets were then utilized to train MLP, Random Forest, and LSTM models for multi-condition health risk prediction.

3.2. Methodology

3.2.1. MLP (Multilayer Perceptron) Working:

A Multilayer Perceptron is a class of feedforward artificial neural network that consists of multiple layers of interconnected nodes. Or we can say it is a type of neural network with layers of connected neurons. It starts with an input layer (raw data), followed by hidden layers that process the data using weights and activation functions, and ends with an output layer for predictions. Each neuron connects to all neurons in the previous layer, using activation functions like ReLU to avoid training issues. The network learns by adjusting weights through backpropagation—correcting errors by moving backward through layers—and uses optimizers like Adam to improve accuracy. Hidden layers help it recognize complex patterns in data. Common activation functions include ReLU (Rectified Linear Unit), which helps mitigate the vanishing gradient problem and accelerates convergence compared to traditional sigmoid or tanh functions.

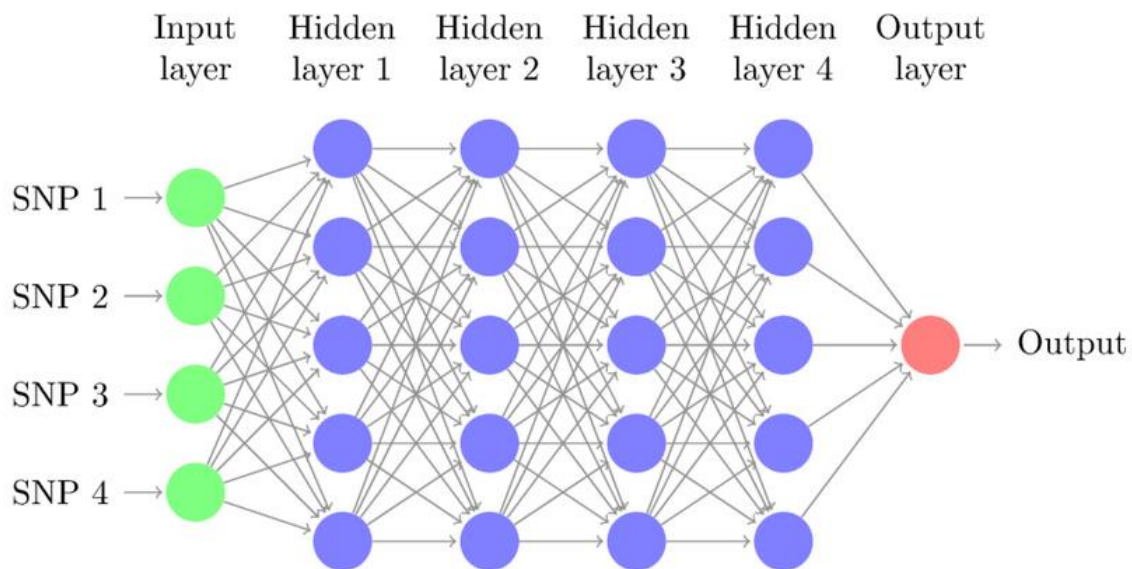


Fig. 3: MLP (Multilayer Perceptron) Working

3.2.2. Random Forest Working:

Random Forest is an ensemble learning method that operates by constructing multiple decision trees during training and outputting the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees. This algorithm builds many decision trees,

each trained on random data subsets and using random features for splits. This randomness reduces overfitting and boost's reliability. During predictions, trees "vote" for the result (classification) or average outputs (regression). It handles messy or high-dimensional data well, works with numbers and categories, and identifies important features without heavy preprocessing.

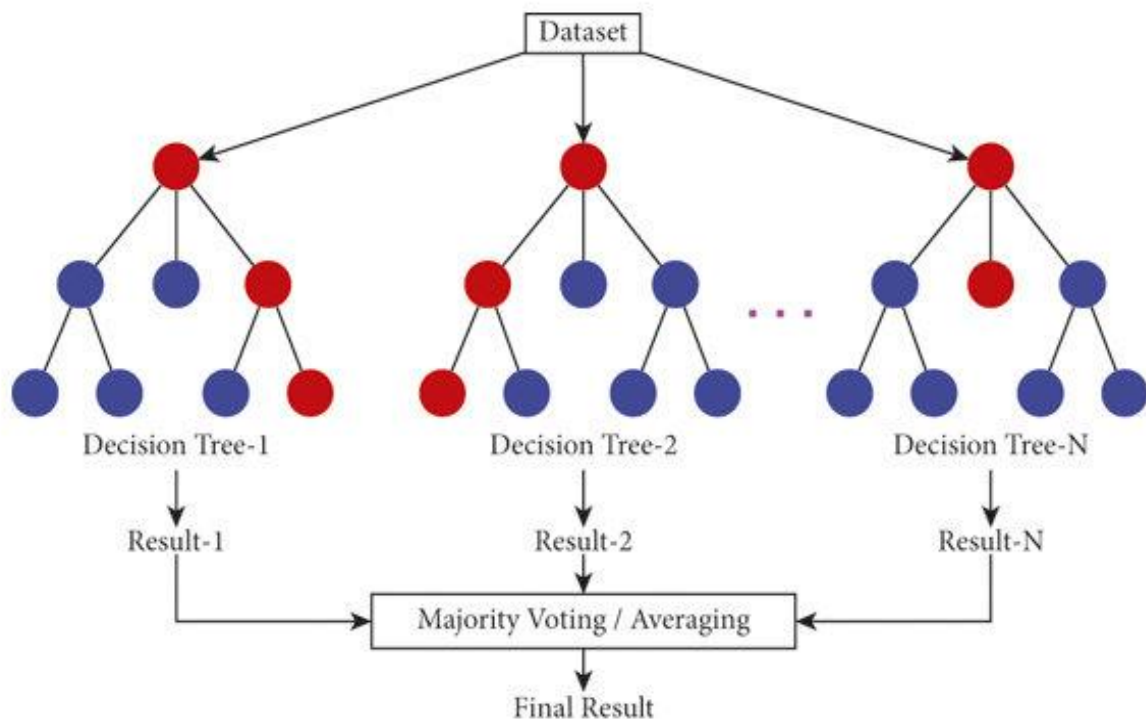


Fig. 4: Random Forest Working

3.2.3. LSTM (Long Short-Term Memory) Working:

LSTMs are specialized neural networks for sequences (e.g., text, time-series). Unlike basic RNNs, they avoid forgetting long-term patterns using "memory cells" managed by three gates:

- **Input gate:** Decides what new info to store.
- **Forget gate:** Removes irrelevant info.
- **Output gate:** Controls what info to pass forward.

These gates use sigmoid functions to regulate data flow, letting LSTMs handle tasks like language translation or stock prediction where context matters over time.

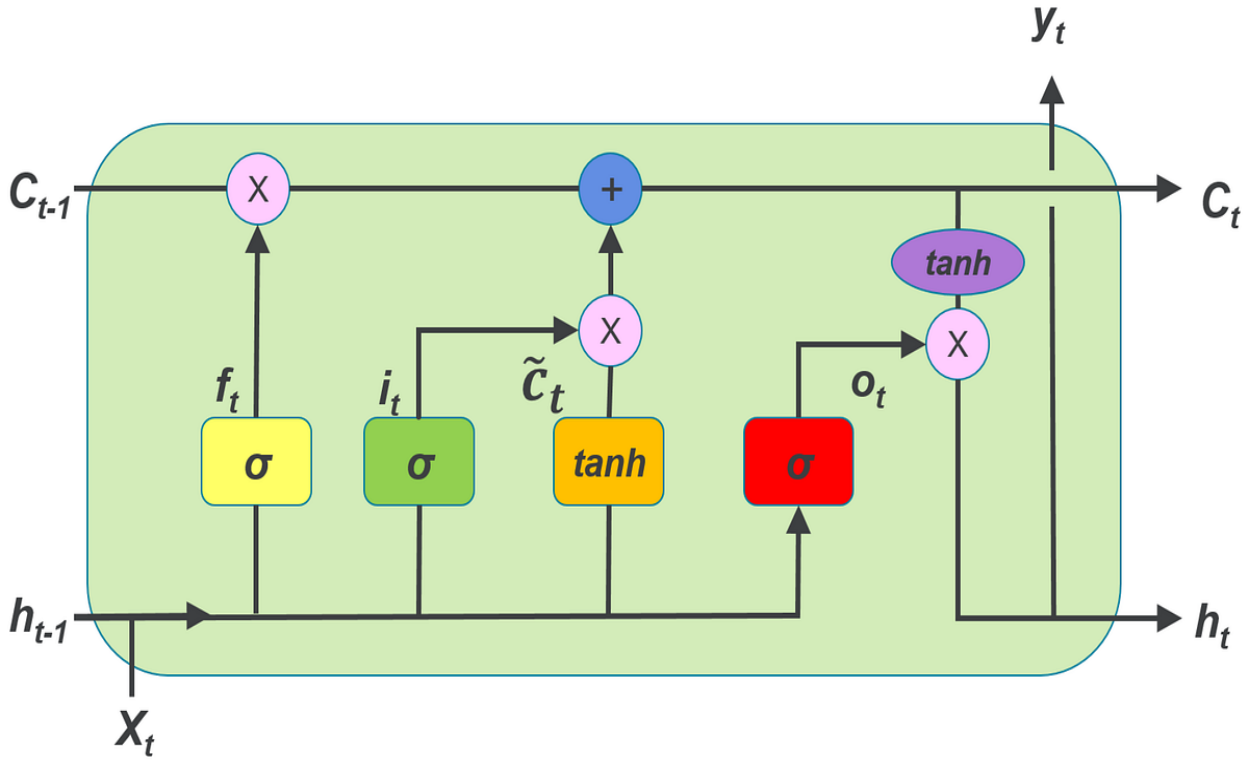


Fig. 5: LSTM (Long Short-Term Memory) Working

3.2.4. MLP Enhancement:

The MLP implementation includes several important enhancements that improve its performance and robustness. First, the data preprocessing stage automatically handles categorical variables through label encoding, ensuring all input features are in numerical format suitable for the neural network. The standardization of features using StandardScaler transforms the data to have zero mean and unit variance, which is particularly important for neural networks as it helps the optimization algorithm converge faster and prevents features with larger scales from dominating the learning process. The MLP architecture itself is carefully configured with two hidden layers containing 100 and 50 neurons respectively, providing sufficient capacity to learn complex patterns without being excessively large. The use of ReLU activation functions helps mitigate the vanishing gradient problem and accelerates training compared to traditional activation functions. The Adam optimizer is employed for its adaptive learning rate capabilities, which often leads to faster convergence than standard stochastic gradient descent. Early stopping is implemented to prevent overfitting by monitoring performance on a validation set and halting training when improvements stagnate, while the validation fraction parameter ensures a proper subset of data is reserved for this purpose. The

fixed random states ensure reproducibility of results, and the `max_iter` parameter prevents excessively long training sessions. Together, these enhancements create a more robust, efficient, and reliable MLP implementation compared to a basic configuration.

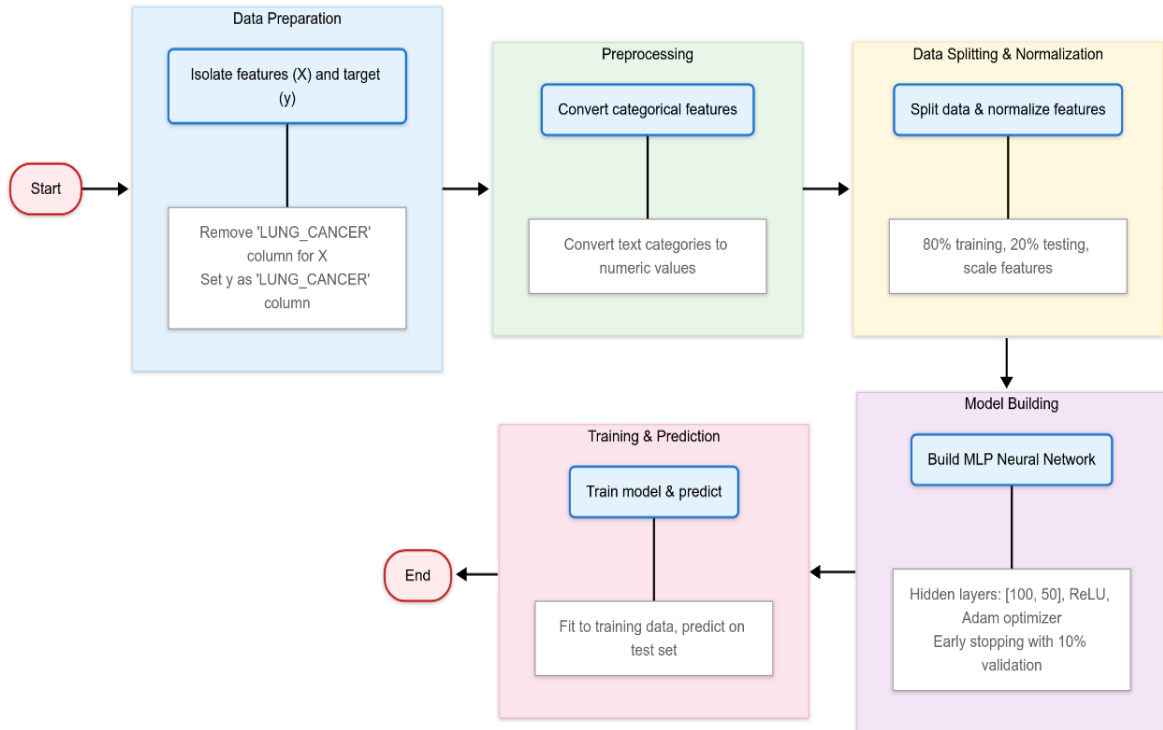


Fig. 6: Flow Chart of the MLP Enhancement algorithm

3.2.5. Algorithm Workflow:

1. Data Preparation:

- Isolate features (X) by removing the 'LUNG_CANCER' column.
- Define the target (y) as the 'LUNG_CANCER' column.

2. Preprocess Features & Target:

- Convert text-based categories in features (X) to numeric labels.
- If the target (y) contains text labels, convert them to numeric values.

3. Split Data:

- Divide the dataset into 80% training and 20% testing subsets.

4. Normalize Features:

- Scale training and testing data to ensure features have consistent ranges.

5. Build MLP Model:

- Create a neural network with:
 - 100 neurons in the first hidden layer, 50 in the second.
 - ReLU activation for nonlinearity.
 - Adam optimizer for weight adjustments.
 - Early stopping to halt training if no improvement (10% validation subset).

6. Train & Predict:

- Fit the model to the training data.
- Use the trained model to predict outcomes for the test set.

4. Results and Discussions

4.1. Lung Cancer Prediction

The performance of various machine learning models was evaluated for lung cancer prediction using F1-score and accuracy. The Random Forest Classifier achieved an F1-score of 0.88 and an accuracy of 0.80, while the baseline Multi-Layer Perceptron (MLP) outperformed it with an F1-score of 0.90 and accuracy of 0.83. The Long Short-Term Memory (LSTM) network showed relatively lower performance (F1-score 0.85, accuracy 0.76), indicating its limitations with non-sequential, static data.

To improve prediction, several feature engineering techniques were introduced. A Risk Score was computed by summing key risk indicators such as allergy, wheezing, alcohol consumption, coughing, and swallowing difficulty. A Smoke-Alcohol interaction feature was added to reflect the compounded risk of smoking and alcohol use. Age was categorized into discrete groups

representing youth, middle-aged, and elderly populations. Furthermore, a Physical Symptoms Count was generated from symptoms like chest pain and shortness of breath, while a Mental Health Score was created by combining anxiety and peer pressure levels.

These engineered features led to notable performance improvements. MLP with the new features yielded an F1-score of 0.91 and accuracy of 0.84. Further extending and refining the feature set by removing redundant attributes produced the best results: an F1-score of 0.94 and accuracy of 0.89. This confirms the importance of domain-specific feature transformation in enhancing model precision and reliability in lung cancer prediction.

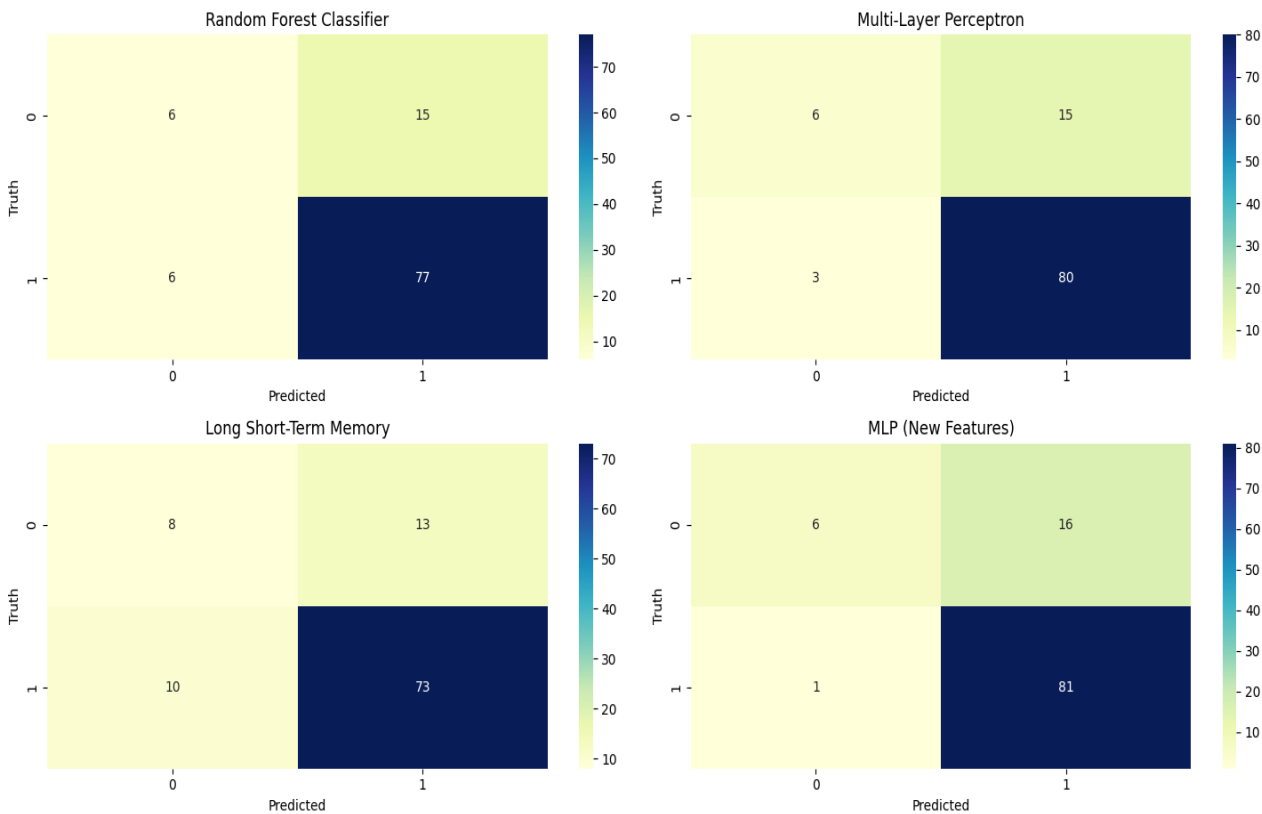


Fig. 7: Confusion Matrix for Lung Cancer Prediction

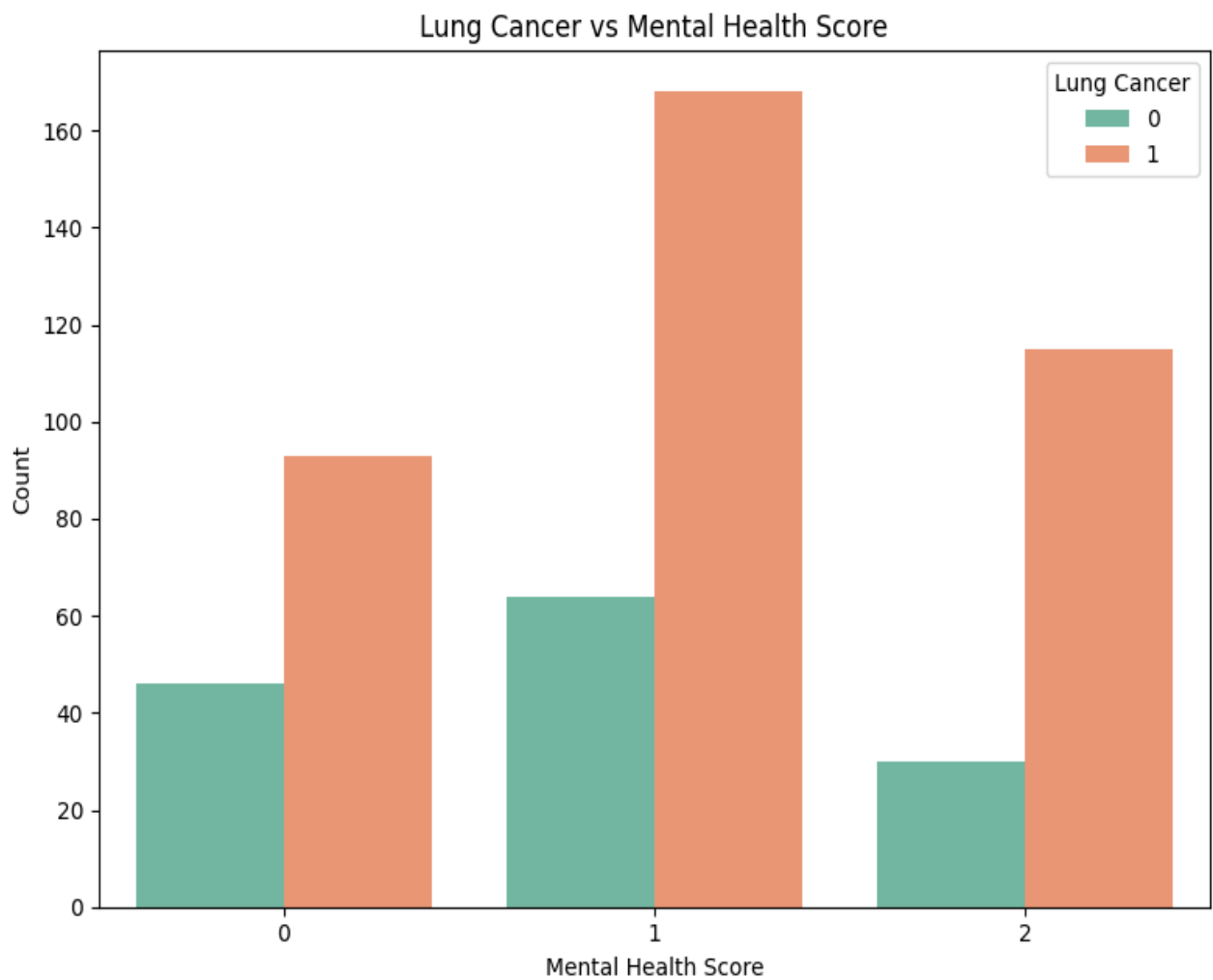
4.2. Mental Health Prediction

To evaluate mental health risk, the original anxiety and peer pressure columns were removed to reduce dimensionality and simulate prediction using indirect features. A derived **Mental Health Score** was previously engineered by summing these two factors and retained for analysis.

Table 2: Mental Health Dataset

GENER	SMOKING	YELLOW_FINGERS	CHRONIC_DISEASE	FATIGUE	ALLEGY	WHEEZING	ALCOHOL_CONSUMING	CUGHING	SHORTNESS_OF_BREATH	SWALLOWING_DIFFICULTY	CHEST_PAIN	LUNG_CANCER	RISK_SCORE	SMOKE_1ALCOHOL	AGE_GROUPS	PHYSICAL_MPTO_MS_COUNT	MENTAL_HEALTH_SCORE
1	0	1	0	1	0	1	1	1	1	1	1	1	4	0	2	4	1
1	1	0	1	1	1	0	0	0	1	1	1	1	2	0	2	3	0
0	0	0	0	1	0	1	0	1	1	0	1	0	2	0	1	3	1
1	1	1	0	0	0	0	1	0	0	1	1	0	2	1	2	2	1
0	0	1	0	0	0	1	0	1	1	0	0	0	2	0	2	2	0

A bar graph comparison between lung cancer presence and mental health score indicated a strong correlation, suggesting that higher mental distress may be associated with increased cancer risk.

**Fig. 8: Relation between Lung Cancer and Mental Health Score**

Prediction was then carried out using this reduced dataset. The **Random Forest** model performed well, achieving an F1-score of 0.81 and an accuracy of 0.78, demonstrating its robustness even with derived variables. In contrast, the **LSTM** model achieved lower results (F1-score 0.66, accuracy 0.64), indicating that sequential modeling did not provide a performance advantage for this task using the given data structure. This confirms that engineered features can preserve predictive power even after dropping direct mental health indicators.

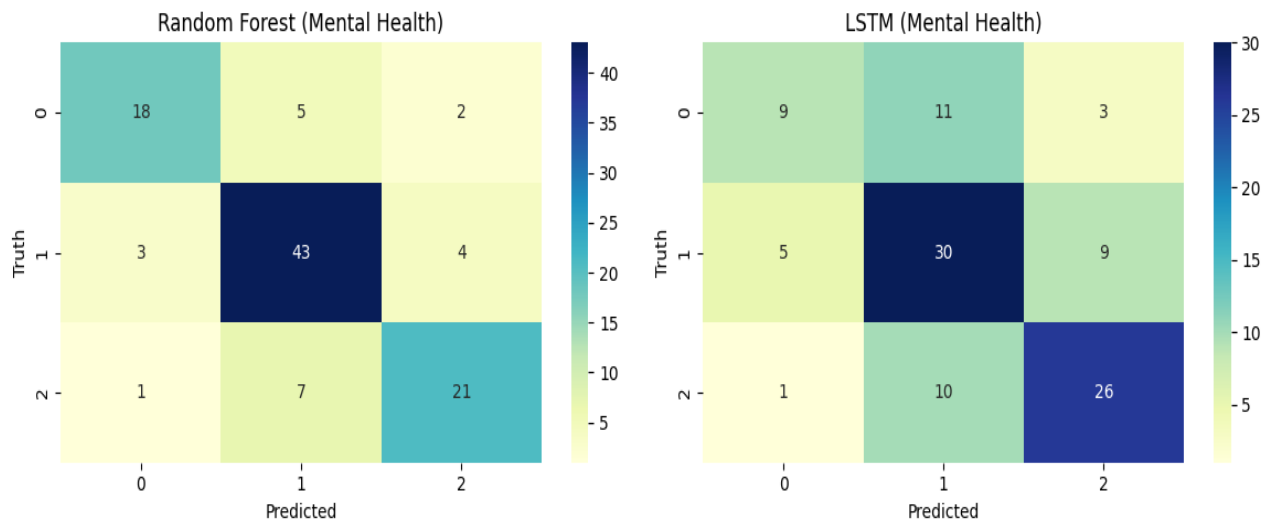


Fig. 9: Confusion Matrix for Mental Health Prediction

4.3. Jaundice Prediction

For jaundice prediction, a **synthetic binary feature** was created by combining three indicators—**alcohol consumption**, **chronic disease**, and **yellow fingers**—assigning a positive jaundice label when at least two were present. After generating this new target, the original three features were excluded to reduce redundancy and potential overfitting.

Table 3: Jaundice Dataset

GEN DER	AGE	SMOK ING	ANXI ETY	PEER_ PRESS URE	FATI GUE	ALLE RGY	WHEE ZING	COUG HING	SHORTN ESS OF BREATH	SWALLOWI NG DIFFICULTY	CHEST PAIN	LUNG_ CANCER	JAUNDICE
1	69	0	1	0	1	0	1	1	1	1	1	1	1
1	74	1	0	0	1	1	0	0	1	1	1	1	0
0	59	0	0	1	1	0	1	1	1	0	1	0	0
1	63	1	1	0	0	0	0	0	0	1	1	0	1
0	63	0	0	0	0	0	1	1	1	0	0	0	0

A bar graph analysis showed a positive correlation between **jaundice presence** and **lung cancer**, suggesting that these conditions may share common risk factors or comorbid pathways.

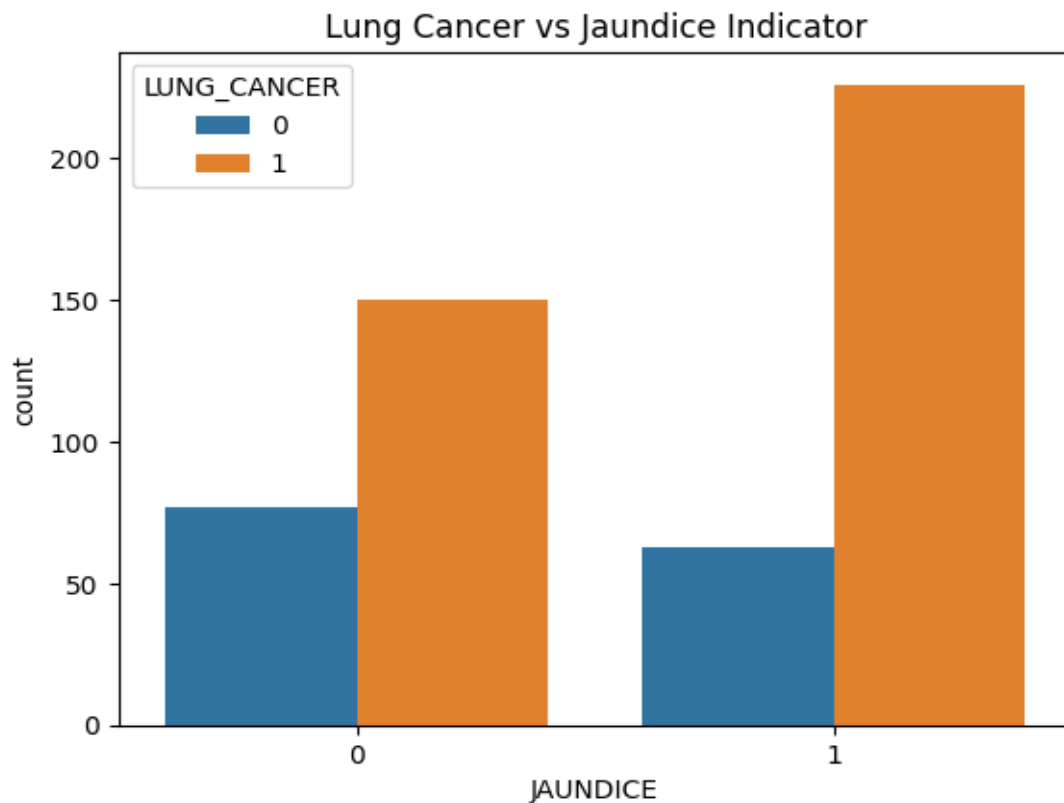


Fig. 10: Relation between Lung Cancer and Jaundice

Prediction results show that the **Random Forest** model outperformed others, achieving an **F1-score of 0.85** and **accuracy of 0.82**, indicating strong classification performance despite using a synthesized feature. The **Multi-Layer Perceptron (MLP)** model followed with a moderate **F1-score of 0.78** and **accuracy of 0.69**, showing slightly lower effectiveness in handling the reduced feature set.

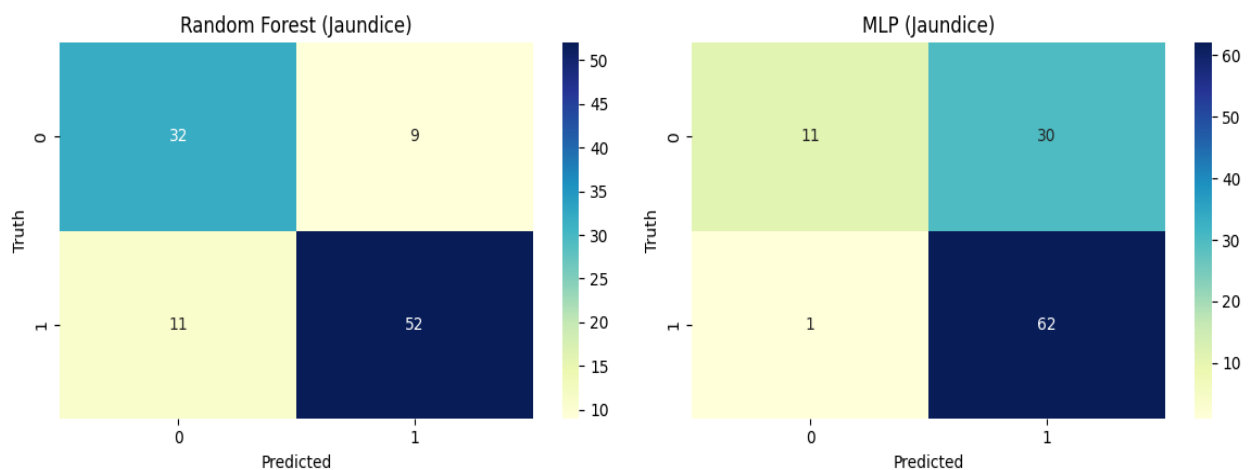


Fig. 11: Confusion Matrix for Jaundice Prediction

4.4. Heart Problem Prediction

The **Heart Problem** label was synthesized by summing multiple key health indicators, including smoking, anxiety, peer pressure, chronic disease, fatigue, allergy, wheezing, alcohol consumption, coughing, chest pain, and shortness of breath. Individuals exceeding a defined threshold were labeled as having potential cardiac issues.

Table 4: Heart Problem Dataset

GE ND ER	SM OK IN G	YELL OW _FI NGE RS	AN XIE TY	PEE R_P RES SUR E	CHR ONI C DISE ASE	FA TI GU E	AL LE RG Y	W HE EZI NG	ALCO HOL CONS UMI NG	CO UG HI NG	SHOR TNES S OF BREA TH	SWAL LOWI NG DIFFI CULT Y	CH ES T PAI N	LUN G_C ANC ER	SM OKE _ ALC OH OL	AG E_ GR OU P	PHYSIC AL_SY MPTO MS_C OUNT	MEN TAL_ HEA LTH_ SCO RE	HEA RT_ PRO BLE M
1	0	1	1	0	0	1	0	1	1	1	1	1	1	1	0	2	4	1	1
1	1	0	0	0	1	1	1	0	0	0	1	1	1	1	0	2	3	0	1
0	0	0	0	1	0	1	0	1	0	1	1	0	1	0	0	1	3	1	1
1	1	1	1	0	0	0	0	0	1	0	0	1	1	0	1	2	2	1	0
0	0	1	0	0	0	0	0	1	0	1	1	0	0	0	0	2	2	0	0

A comparative bar graph analysis illustrated a strong association between **lung cancer** and **heart problem risk**, highlighting the overlapping risk factors between respiratory and cardiovascular conditions.

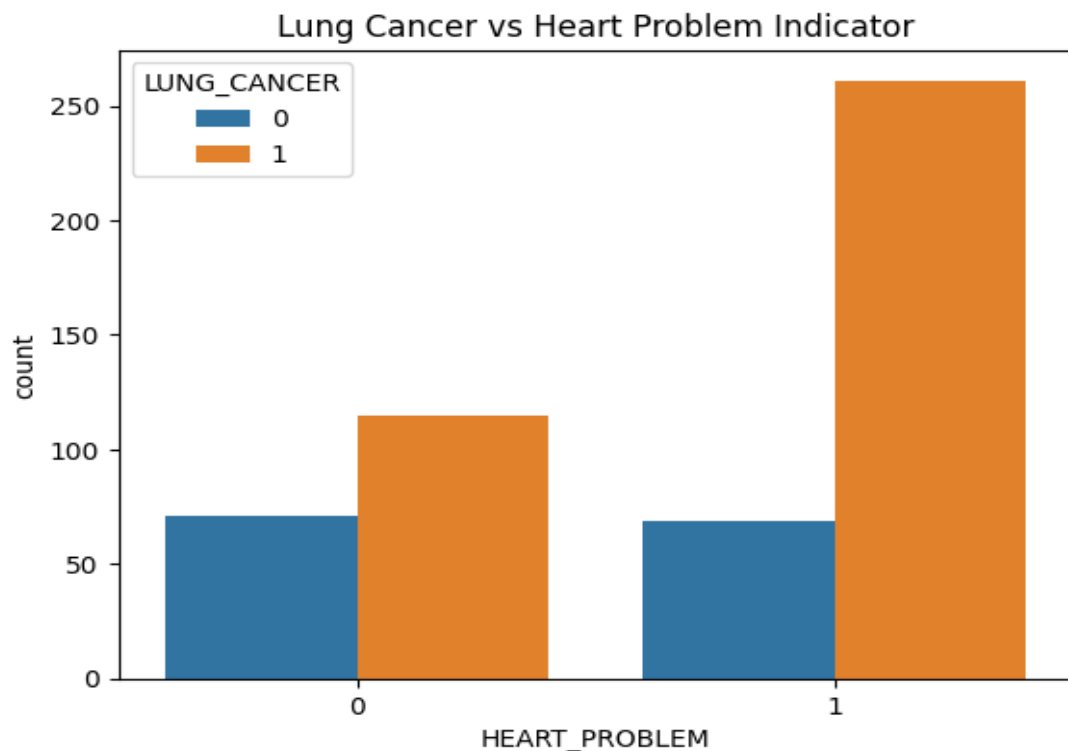


Fig. 12: Relation between Lung Cancer and Heart Problem

Both **Random Forest** and **Multi-Layer Perceptron (MLP)** models achieved high and identical performance, each with an **F1-score of 0.95** and **accuracy of 0.93**, demonstrating the effectiveness of the derived features and model robustness in detecting heart-related complications.

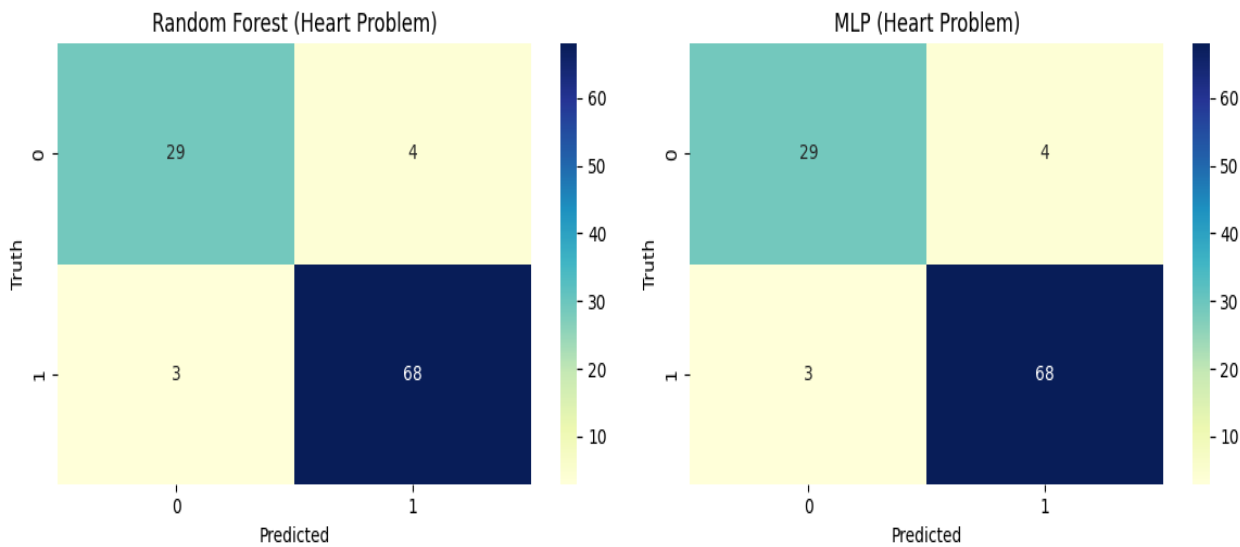


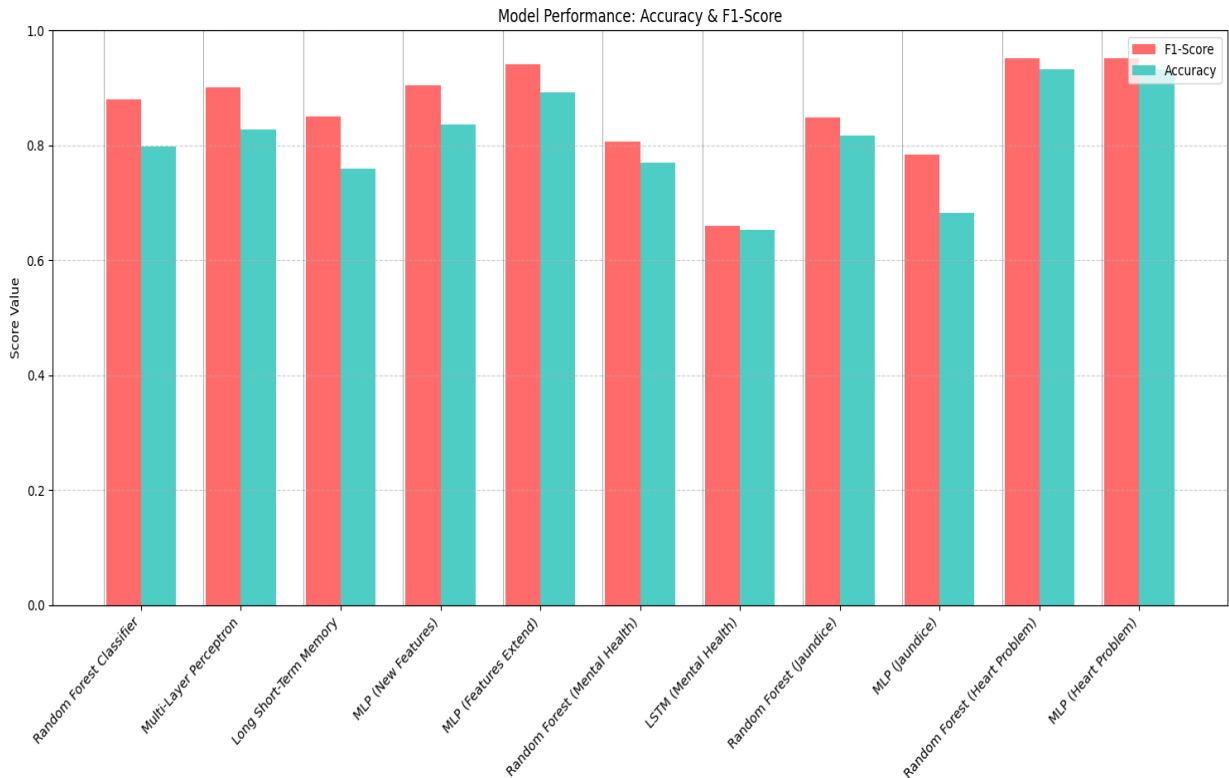
Fig. 13: Confusion Matrix for Heart Problem Prediction

4.5. F1-score and Accuracy

The following presents a compiled summary of each model's F1-score and accuracy through both numerical metrics and visual comparison. The table below summarizes each model’s performance, highlighting F1-scores and accuracy metrics. A bar graph clearly illustrates the performance differences between algorithms, with height variations representing score disparities. This dual presentation format enables quick visual assessment alongside detailed metric evaluation. Together, they provide comprehensive insights into model effectiveness across all prediction tasks.

Table 5: Accuracy and F1 score of all predictions

Model	F1-Score	Accuracy
Random Forest Classifier	0.88	0.80
Multi-Layer Perceptron	0.90	0.83
Long Short-Term Memory	0.85	0.76
MLP (New Features)	0.91	0.84
MLP (Features Extend)	0.94	0.89
Random Forest (Mental Health)	0.81	0.78
LSTM (Mental Health)	0.66	0.64
Random Forest (Jaundice)	0.85	0.82
MLP (Jaundice)	0.78	0.69
Random Forest (Heart Problem)	0.95	0.93
MLP (Heart Problem)	0.95	0.93

**Fig. 14:** Bar graph comparing F1 score and Accuracy of all predictions

5. Conclusion

This study was initiated with the primary objective of developing an accurate, multi-disease prediction model leveraging advanced machine learning and deep learning techniques. The motivation stemmed from the increasing importance of early diagnosis in managing chronic and critical illnesses, particularly in the context of public health systems. Existing research highlighted the potential of models like Random Forest, MLP, and LSTM in disease detection, and this project aimed to build upon such foundations using enriched features and synthetic labels for enhanced prediction capabilities.

Comprehensive data preprocessing and feature engineering were conducted, including encoding categorical variables, creating new features such as Risk Score, Smoke-Alcohol interaction, Age Group, Physical Symptoms Count, and Mental Health Score. Disease-specific labels such as Jaundice and Heart Problem were derived through logical synthesis based on relevant clinical indicators, and original features were pruned to reduce noise and improve model efficiency.

The Lung Cancer Prediction task showed that the use of extended and refined features significantly improved performance, with the MLP (Features Extended) model achieving the highest F1-score of 0.94 and accuracy of 0.89, outperforming traditional MLP, Random Forest, and LSTM models. For Mental Health Prediction, the Random Forest model gave a reasonable performance (F1-score 0.81, accuracy 0.78) even after excluding original mental health indicators, demonstrating the power of engineered features. The Jaundice Prediction model synthesized a new target variable using logical combinations of alcohol consumption, chronic disease, and yellow fingers; Random Forest again led with an F1-score of 0.85 and accuracy of 0.82. For Heart Problem Prediction, both Random Forest and MLP performed equally well with F1-scores of 0.95 and accuracy of 0.93.

In conclusion, the study effectively illustrates how thoughtful feature engineering, along with synthetic target generation and proper model selection, can significantly enhance the predictive power of machine learning models in healthcare. These findings not only contribute to the academic understanding of ML-based diagnostics but also hold practical implications for developing real-time health screening tools in resource-constrained settings.

6. Future Scopes

While this study has made significant progress in developing a multi-disease risk prediction model using enhanced machine learning techniques, several opportunities remain for future research and expansion. First, the prediction of heart problems based on general health features showed high accuracy (F1-Score: 0.95, Accuracy: 0.93). However, this model currently does not incorporate hereditary factors, which are critical in determining cardiac risk. Future work should include family history data to enhance the reliability and personalization of heart disease predictions.

Another area of strength in this study was the extended MLP model for lung cancer prediction, which achieved an F1-Score of 0.94 and Accuracy of 0.89 after carefully engineering and selecting features. Building on this, future studies could investigate organ-specific damage prediction caused by different conditions, utilizing medical imaging and clinical data to model the extent and location of physiological impacts.

Additionally, future models can be developed to perform comprehensive diagnostics based on a complete set of symptoms and medical test results. These models could predict not just one disease but the likelihood of multiple ailments simultaneously, thereby supporting doctors in differential diagnosis. Advancements in natural language processing (NLP) may also enable the integration of clinical notes for richer input data. By broadening data sources, improving personalization, and incorporating predictive insights into specific health impacts, the next generation of models can significantly enhance patient care and public health monitoring.

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 Veer Surendra Sai University of Technology

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